Thoughts And Rambles On Probability And Statistics

Problem Set 1: Just A Bit Of Measure Theory

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These are the end of chapter problems for Chapter 1 of the notes. Equation numbers refer to the full notes.

1. Let $\Omega = \{1, 2, 3, 4\}$. Determine whether or not each of the following is a σ-algebra.

(a) $\mathcal{F}_1 = \{\emptyset, \{1, 2\}, \{3\}, \{3, 4\}, \{1, 2, 3\}, \Omega\}$.
(b) $\mathcal{F}_2 = \{\emptyset, \{3\}, \{4\}, \{2, 3\}, \{1, 3\}, \{1\}, \{1, 2, 3\}, \{1, 2, 3, 4\}\}$.
(c) $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 2, 3\}, \{2\}, \{1, 3, 4\}, \{2, 3\}, \Omega\}$.

2. Let $(\Omega, \mathcal{M}, \lambda)$ be the Lebesgue measure on the interval $[0, 1]$. Let

$$\Omega' = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

Let $\mathcal{F}$ be the collection of all subsets of Ω' of the form $\{(x, y) : x \in A, 0 \leq y \leq 1\}$ for some $A \in \mathcal{M}$. Finally, define a probability P on $\mathcal{F}$ by $P(\{(x, y) : x \in A, 0 \leq y \leq 1\}) = \lambda(A)$.

(a) Prove that $(\Omega', \mathcal{F}, P)$ is a probability triple.
(b) Let P^* be the outer measure corresponding to P and $\mathcal{F}$. Define the subset $S = \{(x, y) : 0 \leq x \leq 1, y = 1/2\}$. (Note that $S \notin \mathcal{F}$.) Prove that $P^*(S) = 1$ and $P^*(S^c) = 1$. What does this say about why S can't be measurable?

3. You have probably heard it before. Now you can solve it rigorously. It is called the "Monty Hall Problem". A prize is placed at random behind one of three doors. You pick a door. To be concrete, let's suppose you always pick door 1. Now Monty Hall chooses one of the other two doors, opens it and shows you that it is empty (or has a donkey). He then gives you the opportunity to keep your door or switch to the other unopened door. Should you stay or switch? Intuition suggests it doesn't matter. The correct answer is that you should switch. Prove it. It will help to specify the sample space and the relevant events carefully. Thus write $\Omega = \{(\omega_1, \omega_2) : \omega_i \in \{1, 2, 3\}\}$ where ω_1 is where the prize is and ω_2 is the door Monty opens.

4. Suppose that a fair coin is tossed repeatedly until both a head and a tail have appeared at least once.

(a) Describe the sample space Ω.
(b) What is the probability that three tosses will be required?

5. Suppose that the events $\{A_n, n \geq 1\}$ are independent. Show that the inclusion-exclusion formula reduces to

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) = 1 - \prod_{k=1}^{n} \left(1 - \mathbb{P}(A_k)\right).$$

6. Show, in the setup of the previous problem, that

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) \geq 1 - \exp\left\{-\sum_{k=1}^{n} \mathbb{P}(A_k)\right\},$$

and that

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty \implies \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = 1.$$

7. Show that the intersection of any collection of σ-algebras on Ω is a σ-algebra, but give an example of two σ-algebras whose union is not.

8. Let λ be the Lebesgue measure on $\mathbb{R}$ and let $A_n = [n, \infty)$. Compute $\lim_n \lambda(A_n)$ and $\lambda(\bigcap_n A_n)$. Which hypothesis of continuity from above fails?

9. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_n = 1) = p_n$ and $\mathbb{P}(X_n = 0) = 1 - p_n$. Show that $X_n \to 0$ almost surely if $\sum_n p_n < \infty$, and that X_n fails to converge almost surely if $\sum_n p_n = \infty$ and $p_n \to 0$.